

Expected Value and Variance (Continuous)

CSU, Chico Math 350

2026-04-20

outline

Expected Value

Expected Value of Functions of Random Variables

Variance and Standard Deviation

References

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Definition

If a random variable has a continuous distribution for which the PDF is f , then the expectation $E(X)$ is defined as:

$$\mu = E(X) = \int_{-\infty}^{\infty} x * f(x) dx$$

Example 1: Jail Time

Suppose X is the random variable that represents the prison sentence in years for persons convicted of grand theft auto and assume that X has a PDF of:

$$f(x) = \frac{1}{9}x^2, \text{ for } 0 < x < 3$$

What is the average length of time these people spend in jail?
Calculate this by hand, and using R.

By hand:

$$E(X) = \int_0^3 x * \frac{1}{9}x^2 = \int_0^3 \frac{1}{9}x^3 = \frac{1}{36}x^4 \Big|_0^3 = \frac{81}{36} = \frac{9}{4}$$

With R:

```
integrand <- function(x) {1/9*x^3}  
integrate(integrand,lower=0,upper=3)  
## 2.25 with absolute error < 2.5e-14
```

The average sentence for grand theft auto is **2.25 years**.

Exercise 1

Find the expected value for the following PDF, both by hand and using R functions:

$$f(x) = 2x, \text{ for } 0 < x < 1$$

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Theorem 4.2

Let X be a continuous random variable and let g be a function.

$$E[g(X)] = \int g(x)f(x)dx$$

The expected value for $g(x)$, a function of X , is the integral of the multiplication of the function of x , times the PDF.

This is the same process we used for discrete cases, despite the different notation here.

Example 2, Pt 1

Let Y have probability density function $f_Y(y) = 2(1 - y)$, $0 \leq y \leq 1$. Suppose that $W = Y^2$, and the PDF of W is:

$$f_W(w) = \frac{1}{\sqrt{w}} - 1, \text{ for } 0 \leq w \leq 1$$

Find $E(W)$ using (A) $f(w)$ directly and (B) using Thm 4.2. Calculate this by hand, and using R.

(A) By hand:

$$\begin{aligned} E(W) &= \int_0^1 w * \left(\frac{1}{\sqrt{w}} - 1 \right) dw = \int_0^1 w^{1/2} - w \, dw \\ &= \frac{2}{3} w^{3/2} \Big|_0^1 - \frac{1}{2} w^2 \Big|_0^1 = \frac{2}{3} - \frac{1}{2} = \frac{1}{6} \end{aligned}$$

With R:

```
integrand <- function(w){w*(1/sqrt(w)-1)}  
integrate(integrand,lower=0,upper=1)  
## 0.1666667 with absolute error < 9.3e-05
```

Example 2, Pt 2

(B) Using $f(y)$:

$$\begin{aligned} E(W) = E(Y^2) &= \int_0^1 y^2 * 2(1 - y) dy = 2 \int_0^1 y^2 - y^3 dw \\ &= 2 \left(\frac{y^3}{3} - \frac{y^4}{4} \right) \Big|_0^1 = \frac{2}{3} - \frac{1}{2} = \frac{1}{6} \end{aligned}$$

With R:

```
integrand <- function(y){y^2*(2*(1-y))}  
integrate(integrand,lower=0,upper=1)  
## 0.1666667 with absolute error < 1.9e-15
```

Exercise 2

A) Suppose that the PDF of a random variable X with a continuous distribution is $f(x) = 2x$ for $0 < x < 1$. Find the expectation of $1/X$. Do this by hand, confirming your results using R.

B) An exam goes poorly for a class, with grades following the distribution:

$$f(y) = \frac{1}{5000}(100 - y), \text{ for } 0 \leq y \leq 100$$

As a way of curving the results, the professor announces that they will replace each person's grade, Y , with a new grade $g(Y)$ where $g(Y) = 10\sqrt{Y}$. Will the professor's strategy be successful in raising the class average above 60? Write down the equation, then use R to calculate the value.

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Definition - Recall Thm 3.9-3.10

Just like for discrete variables, the variance and the standard deviation measures the spread of the random variable around its mean. The formulas remain unchanged for the continuous random variable.

$$\sigma^2 = \text{Var}(X) = E[X^2] - E[X]^2$$

$$\text{Var}(cX) = c^2 \text{Var}(X)$$

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$$

Example 3

Find the variance of the random variable Y , where
 $f_Y(y) = 3(1 - y)^2, 0 < y < 1$.

First find $E(Y)$:

$$\begin{aligned} E(Y) &= \int_0^1 y * 3(1 - y)^2 dy = 3 \int_0^1 y * (1 - 2y + y^2) dy \\ &= 3 \int_0^1 y - 2y^2 + y^3 dy = 3 \left(\frac{y^2}{2} - \frac{2y^3}{3} + \frac{y^4}{4} \right) \Big|_0^1 = \frac{1}{4} \end{aligned}$$

Then find $E(Y^2)$:

$$E(Y^2) = 3 \int_0^1 y^2 * 3(1 - y)^2 dy = 3 \int_0^1 y^2 - 2y^3 + y^4 dy = \frac{1}{10}$$

$$\text{And } \text{Var}(Y) = E[Y^2] - E[Y]^2 = \frac{1}{10} - \frac{1}{16} = \frac{3}{80} = 0.0375.$$

Example 3

Confirming with R functions:

```
f.Ey <- function(y){y*3*(1-y)^2}
f.Eysq <- function(y){y^2*3*(1-y)^2}

(E_Y <- integrate(f.Ey, lower=0, upper = 1))
## 0.25 with absolute error < 2.8e-15

(E_Ysq <- integrate(f.Eysq, lower=0, upper = 1))
## 0.1 with absolute error < 1.1e-15

(Var_Y = E_Ysq$value - (E_Y$value)^2)
## [1] 0.0375
```

Exercise 3

A random variable Y is described by the PDF $f_Y(y) = 2y$ for $0 \leq y \leq 1$. What is the standard deviation of $3Y + 2$?

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references I

Speegle & Clair. Probability, Statistics, and Data: A Fresh Approach Using R, 2021.

Also adapted from notes by Dr. Roualdes, Dr. Donatello, and Dr. Gray.